ENES 246 · DIGITAL LOGIC DESIGN

Lecture 9

Review for Exam 1

Numbers and codes · Boolean algebra and canonical forms · K-maps and the tabular method · NAND and NOR circuits · timed practice set

A	B	F
0	0	0
0	1	1
1	0	1
1	1	0

$$F = A'B + AB' = A \oplus B$$

Reading: Mano & Ciletti, *Digital Design with an Introduction to the Verilog HDL*, 6th ed. — Chapters 1–3 (review)

© 2026 Dr. Stefan Cular. All rights reserved.

Today’s roadmap

1

§1.3

Numbers, codes

Lectures 1–2

2

§2.4

Algebra and forms

Lectures 3–5

3

§3.3

Minimization

Lectures 6 and 8

4

§3.6

Implementation

Lecture 7

By the end of class you can…

- Convert numbers and do signed arithmetic with overflow checks
- Encode with BCD, excess-3, Gray and parity
- Simplify algebraically and write Σ and Π forms
- Minimize with K-maps (SOP and POS) and the tabular method
- Draw NAND-NAND and NOR-NOR circuits

© 2026 Dr. Stefan Cular. All rights reserved.

Agenda for today’s session



Elapsed time	Topic	In-class work
0:00 – 0:05	Welcome; Homeworks 4 and 5 collected	—
0:05 – 0:15	What the exam covers; notation; common errors	—
0:15 – 0:35	Conversions, signed arithmetic, codes	Reviews 1–3
0:35 – 0:50	Simplification, complements, canonical forms	Reviews 4, 5
0:50 – 1:00	Break	
1:00 – 1:25	K-map with don’t-cares, POS, tabular method	Reviews 6–8
1:25 – 1:35	NAND implementation	Review 9
1:35 – 1:50	Timed practice set and solutions	Practice set

© 2026 Dr. Stefan Cular. All rights reserved.

What Exam 1 covers

Lecture	Topic	You should be able to
1	Number systems	convert bases; complements; signed arithmetic; overflow
2	Codes and binary logic	BCD, excess-3, Gray, ASCII, parity; timing diagrams
3	Boolean algebra	prove and simplify; DeMorgan; complements
4	Canonical and standard forms	Σ and Π forms; SOP and POS circuits; the 16 functions
5	Logic families	fan-out, noise margin, power, delay
6	Karnaugh maps	2–4 variables; primes; don’t-cares; POS
7	NAND, NOR, XOR	two-level and multilevel conversion; parity
8	Tabular method	prime implicants; chart; Petrick
10	Combinational design	analysis and design procedure

© 2026 Dr. Stefan Cular. All rights reserved.

Notation used on the exam

Write it this way

Complement: A'. AND: AB. OR: $A + B$. XOR: $A \oplus B$.

Sum of minterms: $F(A, B, C) = \Sigma m(1,4,5)$

Product of maxterms: $F = \Pi M(0,2,3,6,7)$

Don't-cares listed separately: $+ d(\dots)$

The first variable listed is the MSB of the index.

Tag every base: $(247.375)_{10} = (11110111.011)_2$

Common homework errors

Reading division remainders top-down (it is bottom-up).

Forgetting to drop the end carry in 2's-complement subtraction.

Dropping parentheses when complementing.

K-map columns in binary order instead of Gray order.

Missing the wrap-around groups on a 4-variable map.

Leaving a lone literal un-complemented in NAND-NAND.

© 2026 Dr. Stefan Cular. All rights reserved.

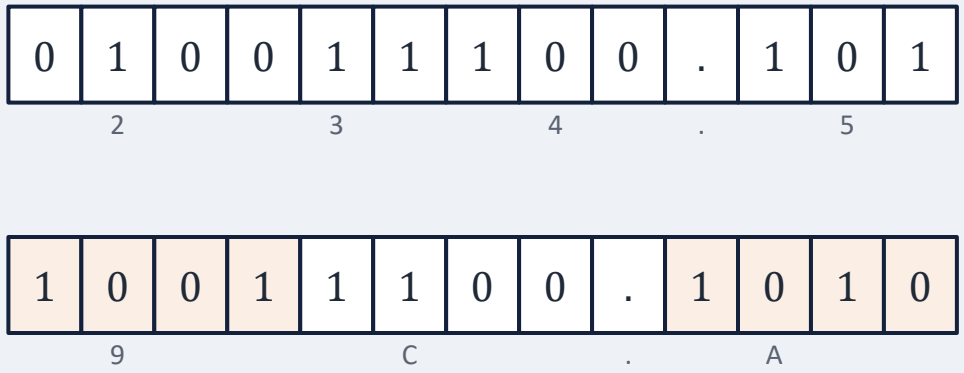
Review 1 — Base conversion

§1.3

REVIEW

Convert $(156.625)_{10}$ to binary, octal and hexadecimal.

Group from the binary point



Octal: groups of 3 (top). Hex: groups of 4 (bottom). Pad the ends with 0s.

1. Largest power of 2 in 156 is 128: $156 - 128 = 28$
2. $28 - 16 = 12$, $12 - 8 = 4$, $4 - 4 = 0$ → 1s at 128, 16, 8, 4: 10011100
3. $0.625 \times 2 = 1.25 \rightarrow 1$; $0.25 \times 2 = 0.5 \rightarrow 0$; $0.5 \times 2 = 1.0 \rightarrow 1$
4. Join: $(156.625)_{10} =$ **$(10011100.101)_2$**
5. Octal, groups of 3: $010\,011\,100\,.\,101 \rightarrow$ **$(234.5)_8$**
6. Hex, groups of 4: $1001\,1100\,.\,1010 \rightarrow$ **$(9C.A)_{16}$**
7. Check: $2 \cdot 64 + 3 \cdot 8 + 4 = 156$; $9 \cdot 16 + 12 = 156$ ✓

© 2026 Dr. Stefan Cular. All rights reserved.

Review 2 — Signed arithmetic

§1.6

REVIEW In 8-bit 2's complement compute (a) $-75 + 40$ and (b) $-75 + (-60)$. Check for overflow.

(a) and (b) column by column

-75	1	0	1	1	0	1	0	1
$+40$	0	0	1	0	1	0	0	0
sum	1	1	0	1	1	1	0	1

-75	1	0	1	1	0	1	0	1
-60	1	1	0	0	0	1	0	0
sum	0	1	1	1	1	0	0	1

1. $75 = 64 + 8 + 2 + 1 = 01001011$; invert, add 1: $-75 =$ **10110101**
2. (a) $+40 = 00101000$. Add: $10110101 + 00101000 = 11011101$
3. Sign 1: negative. Invert 00100010, add 1: $00100011 = 35$
4. So (a) = **−35 ✓** (opposite signs: no overflow)
5. (b) $60 = 00111100$; invert, add 1: $-60 = 11000100$
6. Add: $1\ 01111001$; drop the end carry: $01111001 = +121$
7. Two negatives gave a positive: **overflow**
8. True sum −135 < −128; $V = C_{in} \oplus C_{out} = 0 \oplus 1 = 1$

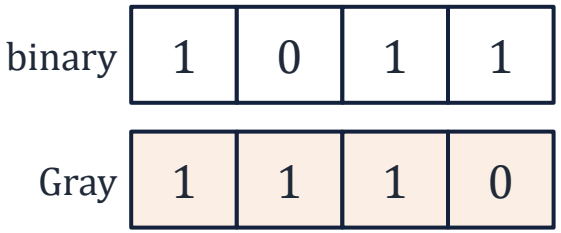
© 2026 Dr. Stefan Cular. All rights reserved.

Review 3 — Codes and parity

§1.7

REVIEW (a) 47 in BCD and excess-3. (b) Gray code of 1011. (c) Even-parity bit for 101 1001.

Code	4	7
BCD	0100	0111
Excess-3	0111	1010



1. (a) BCD: $4 = 0100$, $7 = 0111$ → **0100 0111**
2. Excess-3: $4 + 3 = 7 = 0111$, $7 + 3 = 10 = 1010$ → **0111 1010**
3. (b) Copy the leftmost bit: 1
4. XOR neighbours: $1 \oplus 0 = 1$, $0 \oplus 1 = 1$, $1 \oplus 1 = 0$ → **1110**
5. (c) Count the 1s in 101 1001: four
6. Four is already even: $P = 0$ (odd parity: $P = 1$)

© 2026 Dr. Stefan Cular. All rights reserved.

Review 4 — Algebra and complements

§2.4

REVIEW

(a) Simplify $xy + xy'z + x'yz$. (b) Find the complement of $F = (A + B'C)D'$.

Tools

$x + x'y = x + y$ (use it in both directions)

DeMorgan from the outside in; keep every parenthesis.

Check (a): majority function, $\Sigma m(3,5,6,7)$.

1. (a) $xy + xy'z = x(y + y'z)$ (distributive)
2. $= x(y + z) = xy + xz$ $(x + x'y = x + y)$
3. $xy + x'yz = y(x + x'z)$ (reuse xy: $x = x + x$)
4. $= y(x + z) = xy + yz$ $(x + x'y = x + y)$
5. $F = xy + xz + yz$
6. (b) $F' = (A + B'C)' + (D')'$ (DeMorgan on the AND)
7. $= A'(B'C)' + D$ (DeMorgan, involution)
8. $F' = A'(B + C') + D$ (DeMorgan)

© 2026 Dr. Stefan Cular. All rights reserved.

Review 5 — Canonical forms

§2.6

REVIEW

Write $F = A'B + AC'$ as a sum of minterms and a product of maxterms.

A B C	A'B	AC'	F
0 0 0	0	0	0
0 0 1	0	0	0
0 1 0	1	0	1
0 1 1	1	0	1
1 0 0	0	1	1
1 0 1	0	0	0
1 1 0	0	1	1
1 1 1	0	0	0

1. $A'B$ lacks C: $A'B(C + C') = A'BC' + A'BC$
2. $A'BC' = 010 = m_2$ and $A'BC = 011 = m_3$
3. AC' lacks B: $AC'(B + B') = AB'C' + ABC' = m_4 + m_6$
4. $F = \Sigma m(2,3,4,6)$: the highlighted rows
5. Rows where $F = 0$: 0, 1, 5, 7
6. $$F = \Pi M(0,1,5,7)$$

© 2026 Dr. Stefan Cular. All rights reserved.

Break

10 minutes — we resume at the 1:00 mark

Next: K-maps, POS, the tabular method and NAND circuits

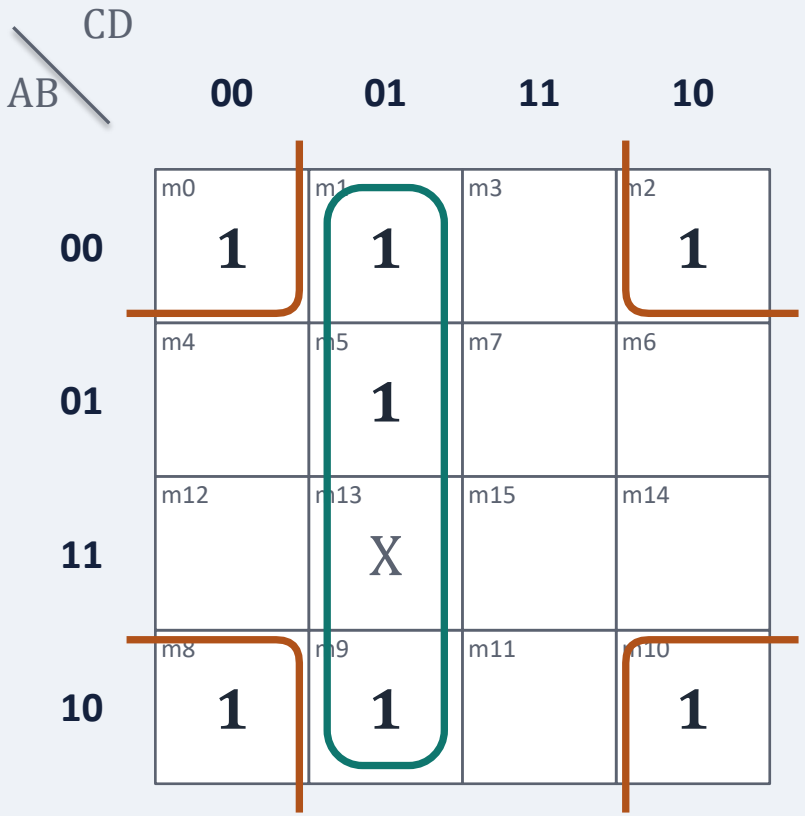
© 2026 Dr. Stefan Cular. All rights reserved.

Review 6 — K-map with don't-cares

§3.5

REVIEW

Minimize $F(A,B,C,D) = \Sigma m(0,1,2,5,8,9,10) + d(13)$ as a sum of products.



1. Place 1s in 0, 1, 2, 5, 8, 9, 10 and an X in 13.
2. Four corners 0, 2, 8, 10: $B = 0$, $D = 0 \rightarrow B'D'$
3. Still uncovered: 1, 5, 9.
4. Treat X at 13 as 1: column $CD = 01$ is a quad.
5. Quad 1, 5, 9, 13: $C = 0$, $D = 1 \rightarrow C'D$
6. Every 1 is covered: $F = B'D' + C'D$, 4 literals

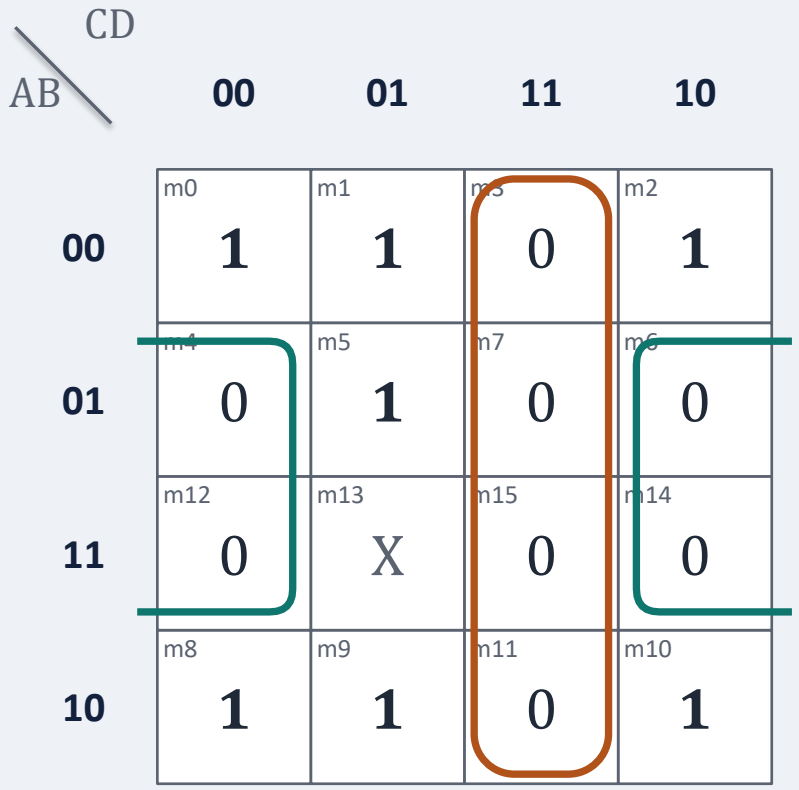
© 2026 Dr. Stefan Cular. All rights reserved.

Review 7 — The same function in POS

§3.4

REVIEW

Write the function of Review 6 as a minimum product of sums.



1. Write 0s in the empty cells: 3, 4, 6, 7, 11, 12, 14, 15.
2. Column $CD = 11$: $C = 1, D = 1 \rightarrow CD$
3. Cells 4, 6, 12, 14: $B = 1, D = 0 \rightarrow BD'$
4. X at 13 stays a 1: it is not grouped with the 0s.
5. So $F' = CD + BD'$
6. DeMorgan: $F = (CD)'(BD')' = (C' + D')(B' + D)$
7. Check against Σm(0,1,2,5,8,9,10) ✓

© 2026 Dr. Stefan Cular. All rights reserved.

Review 8 — Tabular method

REVIEW

Find the prime implicants and a minimum cover of $F = \Sigma m(0,2,3,7,8,10,11,15)$.

Prime	Term	Covers
-0-0	$B'D'$	0, 2, 8, 10
-01-	$B'C$	2, 3, 10, 11
--11	CD	3, 7, 11, 15

1. Group by 1s: 0 | 2, 8 | 3, 10 | 7, 11 | 15
2. Pairs: 0,2 0,8 2,3 2,10 8,10 3,7 3,11 10,11 7,15 11,15
3. Quads: 0,2,8,10 (−0−0); 2,3,10,11 (−01−); 3,7,11,15 (−−11)
4. No quads combine: primes $B'D'$, $B'C$, CD
5. 0 and 8 only in $B'D'$; 7 and 15 only in CD: **essential**
6. Together they cover all eight minterms.
7. $B'C$ is not needed: $F = B'D' + CD$

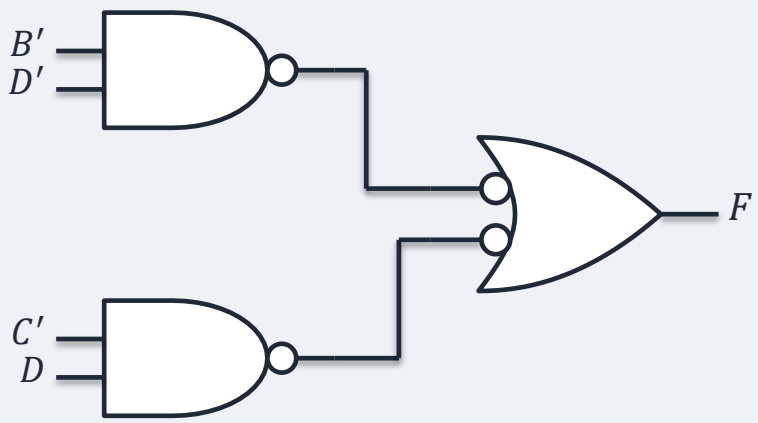
© 2026 Dr. Stefan Cular. All rights reserved.

Review 9 — NAND implementation

§3.6

REVIEW

Implement $F = B'D' + C'D$ (Review 6) with NAND gates only.



1. SOP $B'D' + C'D$: two ANDs and one OR
2. Each AND → NAND
3. The OR → invert-OR (a NAND with input bubbles)
4. Wires: each output bubble meets an input bubble ✓
5. No single-literal term, so no fixes are needed
6. Check: $[(B'D')'(C'D)']' = B'D' + C'D$ ✓ (DeMorgan)
7. NOR-NOR from Review 7: $[(C' + D')' + (B' + D)']'$

© 2026 Dr. Stefan Cular. All rights reserved.

Practice set — 15 minutes

§2.4

YOUR TURN

≈ 15 min

1. Convert $(3AF.8)_{16}$ to decimal.
2. Write -1 and -128 in 8-bit 2's complement.
3. Simplify $(x + y)(x + y') + x'y$.
4. Minimize $F(A, B, C) = \Sigma m(0,2,4,5,6)$ on a map.
5. How many minterms does the 3-variable odd function have?

Work alone, as on the exam.

© 2026 Dr. Stefan Cular. All rights reserved.

Practice set — Solutions 1–3

§2.4

SOLUTION

1. 1. Weights of $3AF.8$: $256,16,1,1/16$

2. $3 \cdot 256 = 768;\ A = 10: 10 \cdot 16 = 160;\ F = 15$

3. $8/16 = 0.5$. Add: $768 + 160 + 15 + 0.5 =$ **943.5**

4. 2. $+1 = 00000001$; invert 11111110, add 1: **11111111**

5. $128 = 10000000$; invert 01111111, add 1: **10000000**

6. 3. $(x + y)(x + y') = x + yy'$ (distributive)

7. $= x + 0 = x$ (complement, identity)

8. So the expression is $x + x'y$

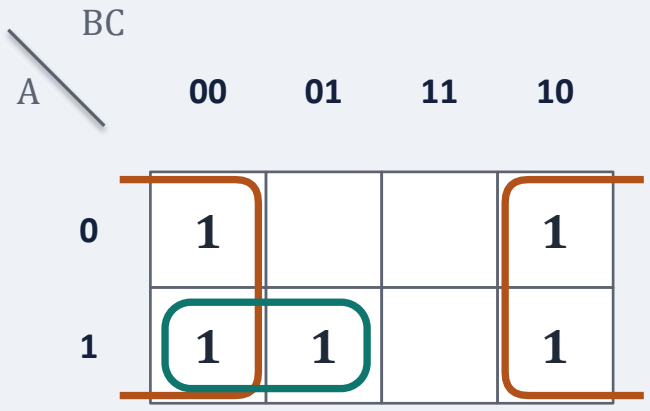
9. $= x + y$ $(x + x'y = x + y)$

© 2026 Dr. Stefan Cular. All rights reserved.

Practice set — Solutions 4–5

§2.4

SOLUTION



1. 4. Place 1s in 0, 2, 4, 5, 6.
2. Outer columns 0, 2, 4, 6 wrap into a quad: only $C = 0 \rightarrow C'$
3. 5 is left: pair 4 (100) with 5 (101): $A = 1, B = 0 \rightarrow AB'$
4. $$F = C' + AB'$$
5. 5. The odd function is 1 when an odd number of inputs is 1:
6. rows 001, 010, 100, 111 → **4** minterms, $\Sigma m(1,2,4,7)$

© 2026 Dr. Stefan Cular. All rights reserved.

Exam 1: how to prepare

Format

Closed book and notes; calculator and pencil.

Show every step: partial credit follows your work.

Check answers by converting back or substituting a row.

Bring

Pencil, eraser, calculator.

Study plan

Redo every Your turn from Lectures 1–8 without the solution.

Rework the homework problems you missed.

Draw 3- and 4-variable maps until the Gray labels are automatic.

Convert one SOP and one POS circuit to NAND and NOR.

Lecture 10 examples are on the exam too.

© 2026 Dr. Stefan Cular. All rights reserved.

Key takeaways

Conversion

divide ↑, multiply ↓

Overflow

$V = C_{in} \oplus C_{out}$

Canonical

Σ and Π: complementary lists

DeMorgan

$(x + y)' = x'y'$

K-map

largest groups, fewest

NAND/NOR

bubbles meet in pairs

Practise until the procedures are automatic.

Homework #6 due at the start of Lecture 13

Next: combinational circuit analysis and design

© 2026 Dr. Stefan Cular. All rights reserved.